



Mahindra University Hyderabad
École Centrale School of Engineering
Minor I Regular Examination
(Batch 2023)

Program: B. Tech

Branch: All (except, CB & BT)

Year: I Semester: I

Subject: Chemistry - I (CH 1101)

Date: 16/09/2023

Start Time: 10.00 AM

Time Duration: 1.5 Hours

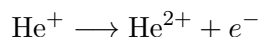
Max. Marks: 30

Instructions

1. Answer all questions.
 2. No clarifications will be entertained during the examination.
 3. If any information is missing, make appropriate assumptions and proceed.
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Q1. (Marks: [7+2+4+1+1=15])

- (i) Derive the expression for radius of Bohr orbit of electron revolving around the nucleus using Bohr's model.
- (ii) The radius of a Bohr orbit of the hydrogen atom is $16 a_0$ (bohr). What is the principle quantum number of the orbit?
- (iii) Using Bohr's model find the ionization energy (in atomic unit, E_h , known as hartree) of helium (He) ion, that is, the minimum energy required for the following process:



- (iv) Energies absorbed or radiated by a blackbody are _____.
- (v) At _____ frequencies classical theory fails to explain blackbody radiation.

Q2. (Marks: [4+7+4=15])

- (i) Write down the expression for time-independent Schrödinger equation; explain its each term.
- (ii) Show that $\Psi(x) = e^{ikx} + e^{-ikx}$ is an eigenfunction of the Hamiltonian operator $\hat{H}$ for a free-particle in one-dimension. What eigenvalue did you obtain? What is the observable here?
- (iii) Which of the following expressions are acceptable wavefunctions, and which are not? State why.

a.

$$\psi(x) = \frac{1}{5-x}, \quad \text{where } 0 \leq x \leq 4$$

b.

$$\psi(x) = e^{x^2}, \quad \text{where } 0 \leq x \leq \infty$$